

The Energy Cost of Privacy in Federated Learning

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Abstract—Differential privacy in federated learning is evaluated along two axes, privacy and utility. On battery-powered edge devices a third decides whether a configuration is deployable at all, and it is not reported. We measure the energy cost of client-level differential privacy with NVML hardware counters differenced around each federated round, sweeping the privacy budget on UCI-HAR, whose clients are the individual study participants, and on CIFAR-10. The per-round cost is power times time, and only one factor transfers: private rounds draw $1.19\times$ the power of non-private ones on both datasets to within 0.5%, while the duration factor differs ($1.27\times$ and $1.13\times$). Neither varies systematically with the budget, so everything budget-dependent acts through the number of rounds — the larger term. Measured as energy-to-target-accuracy, privacy is free at 0.40 accuracy on UCI-HAR, costs $5.7\times$ at 0.70 where only $\epsilon \geq 4$ arrives, and cannot buy 0.85 at any expenditure. Because the cost is paid in joules it inherits the local carbon intensity: an $18.8\times$ range between Algeria’s grid and Norway’s for an identical guarantee, on both datasets. We also report two hazards specific to measuring a privacy mechanism. Deriving noise multipliers inside the timed loop reproduces an entire apparent ϵ trend; and calibrating the noise against *expected* client participation under-charges the clients sampled most, so all 29 of our private runs spend more than their stated budget, by up to 22%.

Index Terms—federated learning, differential privacy, energy efficiency, green AI, carbon footprint, edge computing, human activity recognition

I. INTRODUCTION

Differential privacy in federated learning is evaluated almost exclusively along two axes: a budget ϵ is chosen, the resulting accuracy is reported, and the trade-off between them is the result [1], [2].

A third axis is missing. Adding differential privacy changes how much energy training consumes, and on the battery-powered edge devices federated learning exists to serve, energy decides whether a deployment is possible at all. A privacy configuration a phone cannot afford to run is not a point on the privacy–utility curve; it is off the curve. Green AI research has established that energy and carbon deserve first-class measurement [3], [4], but has concentrated on large centralised training, while federated work on the energy of differential privacy has mostly minimised it analytically under a fixed guarantee [5], [6] or measured it at a single privacy setting [7]. The shape of the cost as ϵ varies — what a designer needs in order to choose a budget — is what we measure here, with NVML hardware counters differenced around each federated round rather than estimated from device power ratings.

The per-round cost is power times time, and only one factor transfers. Private rounds draw $1.19\times$ the power of non-private ones on both datasets, to within 0.5%, despite different models and a fourfold difference in size: the arithmetic of per-sample gradients and clipping, and apparently a hardware constant. The duration factor is $1.27\times$ on UCI-HAR and $1.13\times$ on CIFAR-10. Their product, +51% and +34%, carries no systematic dependence on ϵ , so everything budget-dependent acts through the number of rounds — which an energy-per-round evaluation cannot see.

The cost grows with the accuracy demanded, until it is unbounded. On HAR, private and non-private training cost the same energy to reach 0.40 accuracy, $1.28\text{--}1.79\times$ to reach 0.50, and $5.7\times$ to reach 0.70 — where only $\epsilon \geq 4$ arrives. At 0.85, which the non-private model reaches in 57 rounds, no budget we tested arrives at any expenditure: under client-level accounting the extra training that would close the gap is itself what raises the noise.

The same budget carries very different carbon costs. The privacy premium on 1000 HAR runs is $106\text{--}148\text{ gCO}_2\text{eq}$ on Algeria’s gas-dominated grid against $5.7\text{--}7.9\text{ g}$ in Norway, an $18.8\times$ range the privacy–utility framing cannot express.

UCI-HAR [8] is our primary dataset: its clients are the individual volunteers the data came from, so the client boundary is a real person rather than a synthetic partition. CIFAR-10 carries the results needing a harder task.

Measuring a privacy mechanism also exposes two hazards, and we report both because any such study faces them. Deriving the noise multiplier costs more for a loose budget than a tight one, so run inside the federated loop it lands in the measurement window and takes the shape of a cost of privacy. And the reported ϵ is not the spent one: client sampling is binomial, so the clients participating most overspend the budget the accountant was paid for — in all 29 of our private runs, by up to 22%.

II. RELATED WORK

A. Measuring the energy of training

Green AI made energy and carbon reportable quantities rather than externalities [3], [4], and tools now estimate them from utilisation traces and grid factors [9]. Comparisons against instrumented hardware find those estimates diverge substantially from measurement [10], which is why we read NVML counters instead. Patterson et al. [11] established that where a job runs changes its emissions by $5\text{--}10\times$; our

geographic result applies that to a parameter rather than to a schedule. Savazzi et al. [12] give an analytical energy and carbon framework for federated learning, but without a privacy mechanism.

B. Energy and differential privacy in federated learning

One line of work treats DP noise as a constraint inside a wireless resource-allocation problem and minimises device energy under it, either by harvesting channel noise as the privacy mechanism [6], [13] or by co-designing compression with the noise budget [5]. Energy there is an optimisation objective evaluated through a convergence bound; it is not measured, and ε is a constraint rather than a swept variable. Closest to us empirically, Schwermer et al. [7] built a physically distributed edge testbed and measured traffic, energy and training time for a CNN on MNIST with DP enabled, finding software estimates deviating by up to 35%. Their contrast is DP on versus off at one setting; ours is the shape of the cost across the budget. Aroussi et al. [14] schedule ε adaptively on Raspberry Pi and ESP32 nodes for a 20–25% energy gain — the gain of a controller, not the cost curve a designer needs to choose the budget.

C. Rounds as the mechanism

That the cost should act through the round count is known analytically. Wei et al. [15] prove that under client-level DP there is an optimal number of aggregations for a given protection level, beyond which convergence degrades, and later derive a method to find it [16]; the same optimum reappears for personalised federated learning and in control formulations adapting noise and rounds jointly [17]. These results say where to stop but not what continuing costs, the round being an abstract unit in them. We price it, and find the optimum resolvable only where the budget is tight enough for the decline to clear the measurement noise.

D. Our position

We therefore contribute the missing empirical axis: hardware-counter energy swept across ε under whole-run client-level accounting [1], [18], decomposed into a per-round overhead that carries no systematic budget dependence and a convergence penalty that carries all of it, and reported as energy-to-target-accuracy rather than energy per round or energy per run. Two of our findings are about measuring this at all rather than about the result, and neither appears in the work above: an instrument attached to a privacy mechanism scales with the privacy parameter, and calibrating client-level noise against expected participation under-charges the clients that happen to be sampled most.

III. METHODOLOGY

A. Accounting Model

We enforce *client-level local DP* with an untrusted server: each client holds its own (ε, δ) budget over the entire federated run, accounted over every local SGD step it takes in every

round it joins [18] — not per round, and not per local epoch. For a client with n local training examples,

$$s_{\text{total}} = R \cdot q_c \cdot E \cdot \lceil n/B \rceil \quad (1)$$

with R rounds, client sampling fraction q_c , E local epochs and batch size B . We invert the RDP accountant [19] once, before training, to find the noise multiplier σ that spends ε at δ after s_{total} steps, and hold σ constant for the run.

We do not use the privacy engine’s running accountant: in simulation each client is rebuilt every round, so an engine-internal accountant resets every round and reports a per-round ε while the true budget compounds. We claim no amplification by client subsampling, which would need secure aggregation or a shuffler. The clipping norm C is fixed across all budgets, since tuning it per ε would confound the comparison.

B. What the Reported ε Actually Bounds

Equation 1 uses the *expected* number of rounds a client joins, $R \cdot q_c$. The realised number is a binomial draw around it: at $R = 100$ and $q_c = 0.5$ the expectation is 50 with a standard deviation of 5, so a client sampled 60 times takes 20% more steps than the accountant was paid for. The assurance that a reported ε is an upper bound rests on claiming no subsampling amplification and says nothing about over-participation.

We therefore recompute each client’s spent budget afterwards, from the participation logged every round, partition sizes rebuilt deterministically, and the same accountant. Participation ranged over 33–64 rounds on HAR and 8–22 on CIFAR-10, and **all 29 private runs exceed their stated budget**, by 4.1% to 22.3% (Table I). A study reporting client-level DP in simulation should either report the worst realised budget, as we do, or calibrate σ against a participation bound.

C. Measuring Energy

GPU energy comes from NVML’s `nvmlDeviceGetTotalEnergyConsumption`, a hardware counter in accumulated millijoules [20], differenced around each round: measured rather than modelled from a TDP figure, and free of the integration error of power sampling. The window runs from dispatch of the round’s training messages to the end of aggregation, covering everything the round costs on the device. An idle baseline probed over a quiet five-second window, after waiting for the board to settle, is reported and subtracted — 17.93 W (sd 0.12) on the HAR sweep — over the window the counter actually covers rather than over whole-process wall-clock, which would over-subtract by roughly 40%.

Deriving σ is kept out of that window entirely. Inverting the accountant binary-searches it, and the search costs far more for a loose budget than a tight one — 0.086 s at $\varepsilon = 0.5$ against 0.693 s at $\varepsilon = 8$. Called once per client per round inside the measured region, as the natural implementation does, this alone reproduced an entire wall-clock trend across ε while executed step counts stayed flat: an accountant artefact shaped exactly like a cost of privacy. We derive every σ the sweep needs before any measurement begins.

D. Verification Before Interpretation

The energy axis means nothing unless every budget ran the same workload, so a gate runs before any number is quoted. It checks that each condition has all its seeds; that wall-clock is flat across the private conditions; that executed optimiser steps match the count the accountant was calibrated against; that every client trained on the GPU the counter is watching, since a silent fallback to CPU would record the energy of an idle card; and that idle baselines agree across runs, which caught one probed while the board was still busy at 300 W. Both floors are computed from the across-seed spread rather than assumed. Run order is randomised so drift over a sweep can be told apart from a budget effect: on HAR, wall-clock correlates with execution order at $\rho = -0.154$ and with ε at $\rho = -0.098$, the two predictors themselves decorrelated at $\rho = -0.196$.

E. Setup

UCI-HAR [8] gives 21 clients, one per training participant of the published subject-disjoint split; the nine test participants are held out, so central accuracy measures generalisation to people the federation never saw. Each client splits its windows 80/20 *within each activity class* — necessary because the activities are recorded in blocks and the windows overlap by 50%, so an ordered cut leaves clients training on a subset of the classes. The model is a three-block 1D CNN over the nine raw inertial channels (37k parameters, no batch normalisation, which per-sample gradients forbid). CIFAR-10 gives 20 clients partitioned by Dirichlet($\alpha = 0.5$) [21] and a 62k-parameter CNN.

Both sweeps use $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ with three seeds, $\delta = 10^{-5}$, $q_c = 0.5$, $B = 32$, learning rate 0.01, one local epoch and $C = 1.0$; HAR runs 100 rounds and CIFAR-10 runs 30, simulated with Flower [22] on one NVIDIA RTX 5080 (PyTorch 2.10, CUDA 12.8). We do not enable deterministic kernels: they would change which kernels run and therefore what we measure, so reproducibility comes from averaging seeds.

IV. RESULTS

Every figure is a mean over three seeds. Two resolutions bound what we claim: the energy floor, the worst within-condition across-seed spread of board power, is ± 4.89 W (7.80%) on HAR and ± 5.94 W (8.91%) on CIFAR-10; the accuracy floor, the same statistic on final accuracy, is ± 0.051 on HAR and ± 0.031 on CIFAR-10. No smaller difference is reported as an effect.

The HAR sweep passes every gate: 18/18 runs, wall-clock flat to 0.3% across the private conditions against 38.0 s without privacy, and 89 optimiser steps per round throughout. On CIFAR-10 one $\varepsilon = 4$ run was measured a month after the others and is excluded for breaking the flatness gate; including it moves the mean overhead from +34% to +36%, inside the floor either way, so that condition is reported on two seeds.

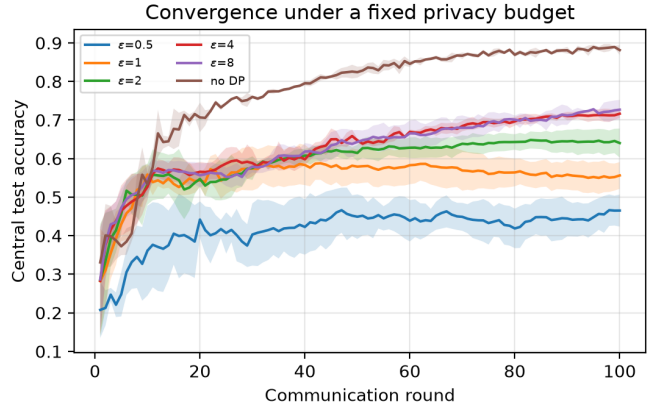


Fig. 1. Central test accuracy against communication round on UCI-HAR over 100 rounds, means of three seeds with ± 1 SD bands. No stopping marks are drawn because none is resolvable here: the largest peak-to-final drop is 0.032 at $\varepsilon = 1$ against a ± 0.042 across-seed spread. What the figure does show is the quality ceiling — the non-private model reaches 0.88 while $\varepsilon = 0.5$ plateaus near 0.47, and no amount of further training closes that gap, because the training itself spends budget.

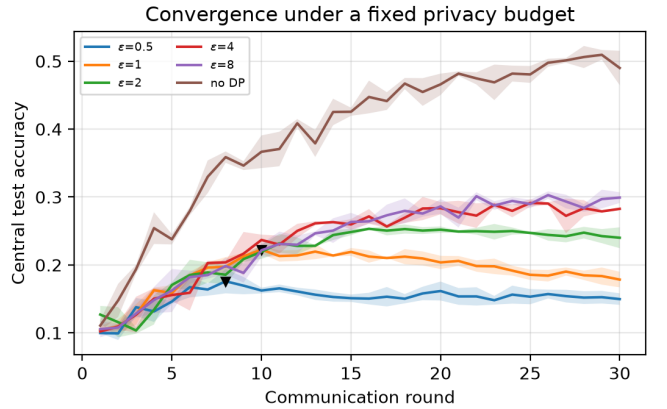


Fig. 2. The same curves on CIFAR-10 over 30 rounds, where a stopping round is resolvable under a tight budget. Markers show the energy-optimal stopping round, drawn only where the peak-to-final drop exceeds the across-seed spread of final accuracy: at $\varepsilon = 0.5$ (round 8, all three seeds peaking within one round of each other) and $\varepsilon = 1$ (rounds 10–16). At $\varepsilon \geq 2$ and without privacy no decline clears the spread, so no marker is drawn. The rounds after the $\varepsilon = 0.5$ peak account for 60–68% of the run’s net energy and buy negative accuracy.

A. The Per-Round Cost Is Power Times Time

Net of idle, private training costs 21.1 J per round on HAR against 14.0 J, and 107.9 J against 80.4 J on CIFAR-10 — +51% and +34% (Table I). Two numbers that far apart look like an unstable measurement until they are decomposed. Private rounds draw more power *and* last longer, and the two factors behave differently:

	Power	Duration	Energy
UCI-HAR	1.190×	1.267×	1.507×
CIFAR-10	1.196×	1.132×	1.354×

The power factor is the same to within 0.5% across two datasets with different models, different input shapes and a 4×

TABLE I
PER-ROUND ENERGY ON UCI-HAR (100 ROUNDS, 21 CLIENTS), MEANS OVER THREE SEEDS. “WORST ε ” IS THE LARGEST BUDGET ANY SINGLE CLIENT ACTUALLY SPENT, RECOMPUTED FROM LOGGED PARTICIPATION (SECTION III-B). CIFAR-10 σ AT 30 ROUNDS IN BRACKETS

ε	Gross (W)	Net (W)	Net (J/rd)	Overhead	Worst ε	σ range
0.5	59.66	42.13	20.3	+45.4%	0.570	17.50–20.62 [2.77–6.09]
1	59.27	41.60	20.0	+43.4%	1.122	9.22–10.94 [1.59–3.32]
2	61.13	43.01	20.7	+48.3%	2.285	4.94–5.90 [1.04–1.90]
4	65.77	47.18	22.7	+62.2%	4.250	2.75–3.25 [0.78–1.20]
8	62.68	44.71	21.5	+54.0%	9.112	1.64–1.90 [0.62–0.85]
∞	54.47	36.75	14.0	—	—	—

difference in model size: the arithmetic of per-sample gradients and clipping, apparently a hardware constant. The time factor is not, depending on how the workload fills the device.

Neither orders with ε . The overhead runs +45.4%, +43.4%, +48.3%, +62.2%, +54.0% on HAR and +36.0%, +38.6%, +30.7%, +30.6%, +35.6% on CIFAR-10, non-monotone in both. On CIFAR-10 the spread across budgets is 6.0% of the mean, inside the 8.91% floor, so the per-round cost is ε -independent within resolution; on HAR it is 12.5% against 7.80%, not resolvably flat, but being non-monotone it is scatter above our resolution rather than a budget effect. Either way the per-round cost carries no systematic budget dependence, which is what the rest of this section rests on.

B. The Cost Grows with the Accuracy Demanded

Measured against the quality a deployment wants, the cost is not flat at all (Table II, Fig. 1). On HAR, reaching 0.40 accuracy costs 185–357 J net whether or not privacy is on: at this quality privacy is free. At 0.50 the private conditions cost 1.28–1.79 $\times$ the non-private 197 J. At 0.70 only $\varepsilon \geq 4$ arrives at all, at 1710 J against 301 J, a factor of 5.7. At 0.85, which the non-private model reaches in 57.0 rounds and 839 J, no budget we tested arrives at any expenditure.

CIFAR-10 shows the same shape at its own scale: 0.15 costs 644–760 J against 611 J, 0.20 costs 1.40–1.63 $\times$ the non-private 677 J, and 0.25 is reached only at $\varepsilon \geq 2$, at 1.96–2.29 $\times$.

Unreachability is the sharpest form of the cost: above a budget-dependent ceiling the accuracy cannot be bought at all, because the extra training that would close the gap is itself what raises σ .

C. A Stopping Round Appears Only Under a Tight Budget

Under a tight enough budget the accuracy curve peaks and declines, so training past the peak spends energy and loses accuracy at once. Whether a peak is resolvable is a question about noise rather than about the argmax, so we mark a stopping round only where the peak-to-final drop exceeds the across-seed spread of final accuracy in that condition.

Two CIFAR-10 budgets qualify (Fig. 2). At $\varepsilon = 0.5$ the mean curve peaks at 0.176 at round 8 and ends at 0.150 — a drop of 0.026 against a ± 0.010 spread, the three seeds peaking at rounds 8, 8 and 9 — and the rounds after the peak account for 60–68% of the run’s net energy. At $\varepsilon = 1$ the drop is 0.044 against ± 0.014 , the seeds peak between rounds 10 and 16,

and 42–61% of the energy falls after it. At $\varepsilon \geq 2$, and without privacy, no decline clears the spread within 30 rounds.

UCI-HAR produces no resolvable stopping round at any budget: its largest drop is 0.032 at $\varepsilon = 1$ against a ± 0.042 spread, and at $\varepsilon = 0.5$ its seeds peak at rounds 16, 54 and 65 while the mean curve peaks at round 98 — an argmax wandering over a plateau, not a decline. With σ near 20 the model never climbs far enough to have anything to give back. The effect is therefore real but conditional, with the threshold between $\varepsilon = 1$ and $\varepsilon = 2$ on CIFAR-10.

D. Under a Tight Budget, Tuning Stops Working — On One Task

We vary one factor at a time at $\varepsilon \in \{1, \infty\}$ with three seeds per cell, reading every difference against floors measured within a cell: ± 0.041 on accuracy for HAR, ± 0.024 for CIFAR-10.

On HAR the private model is noise-limited. At $\varepsilon = 1$ final accuracy spans 0.037 across one, two and five local epochs and 0.018 across $C \in \{0.5, 1, 2\}$, both inside the floor and both non-monotone. **On CIFAR-10 the same two knobs are decisive**, falling monotonically from 0.182 to 0.114 to 0.100 across the epoch axis (3.4 $\times$ the floor) and from 0.213 to 0.182 to 0.100 across the clipping axis (4.7 $\times$). The identical experiment is inert on one task and dominant on the other, and the floors say why: CIFAR-10 resolves accuracy twice as finely while its private model retains more signal to lose.

The standard efficiency lever stops paying. More local work per round is how federated learning ordinarily buys fewer rounds, and without privacy it works: on HAR the energy to reach 0.70 falls from 345 J at one local epoch to 193 J at five. Under $\varepsilon = 1$ the target is reached at no epoch count while energy per round rises from 22 J to 49 J, and on CIFAR-10 it is worse than useless — accuracy collapses while post-peak waste grows from 1785 J to 7312 J. Local steps enter s_{total} , so extra local work forces a larger σ and cancels the convergence gain it bought.

Clipping is free in compute. C leaves energy per round unchanged (22 J at all three values on HAR, 112–119 J on CIFAR-10), so it acts purely through convergence, in a way final accuracy alone hides: $C = 2$ peaks at 0.187 at round 7 and ends at 0.100, giving back 46.7% of its peak against 2.7% at $C = 0.5$. Our fixed $C = 1.0$ is conservative rather than optimal.

TABLE II
ENERGY TO REACH A TARGET ACCURACY ON UCI-HAR (J NET OF IDLE;
ROUNDS IN PARENTHESES). “—” MARKS TARGETS NOT REACHED WITHIN
100 ROUNDS. ‡: 2 OF 3 SEEDS

ϵ	0.40	0.50	0.70	0.85
0.5	357 (13.0)	334‡ (12.0)	—	—
1	201 (4.7)	258 (7.7)	—	—
2	197 (4.0)	353 (12.0)	—	—
4	210 (4.3)	253 (6.3)	1710 (74.3)	—
8	187 (3.3)	269 (7.7)	1699 (76.3)	—
∞	185 (6.0)	197 (7.0)	301 (15.0)	839 (57.0)

TABLE III
GCO₂EQ PER 1000 UCI-HAR TRAINING RUNS. “PRIVACY” IS THE
PRIVATE MINUS NON-PRIVATE DIFFERENCE ACROSS ϵ

Grid	DP	No DP	Privacy
Norway	21	15	6–8
France	46	32	12–17
Algeria	403	281	106–148
India	521	363	137–191

E. Carbon Cost by Grid

Table III scales the measured HAR energy to 1000 runs and apply published grid carbon intensities, repeating the measured workload rather than extrapolating to other hardware. The cost attributable to privacy runs from 106–148 gCO₂eq on Algeria’s gas-dominated grid to 5.7–7.9 g in Norway, a factor of 18.8 for the same computation and the same guarantee. CIFAR-10 gives 115–146 g against 6.1–7.8 g, the same premium despite a workload five times larger per round. Selecting a privacy budget is therefore partly a geographic decision, which the privacy–utility framing cannot express.

V. DISCUSSION

A. Two Costs, Two Remedies

The per-round overhead is an implementation cost, and the decomposition says which part. The power factor is the same on both datasets, so it is the arithmetic of per-sample gradients and clipping and will yield only to better kernels or amortised clipping; the time factor is not, so it is where a scheduling or batching change could act. Neither varies systematically with the budget, and everything that does acts through the round count, bounded by the privacy accounting rather than by any implementation. An evaluation reporting energy per round sees only the first and concludes that privacy costs a third to a half again; measured against the accuracy a deployment needs, the second is several times larger and sometimes infinite. The third axis therefore does not merely rank configurations, it changes which engineering effort is worth spending.

B. Privacy Sets a Quality Ceiling

The cost of privacy is not a constant multiplier: on HAR it is invisible at 0.40 accuracy, $5.7\times$ at 0.70, and unbounded at 0.85. Past the ceiling a budget implies, the trade is not energy for accuracy but nothing for nothing. The usual formulation — choose ϵ , then see what utility follows — treats the two as sequential; our results suggest choosing them jointly, since a utility-only evaluation reports a lower number without

saying whether more computation would fix it. It would not: the round budget is part of the privacy budget, so training to convergence is self-defeating, the extension being what prevents convergence.

C. When Tuning Works Is Itself a Finding

The same two-knob experiment is inert on HAR and decisive on CIFAR-10, and that contrast is more useful than either result alone: whether hyperparameter search is worth its energy under DP depends on whether the private model still retains signal to lose. Where it does not, the levers that remain are the privacy and round budgets. And increasing local work per round — the standard way to buy fewer rounds — stops paying on HAR and actively harms on CIFAR-10, because local steps enter s_{total} and force a larger σ .

D. Where the Computation Runs

Measured in joules, the cost inherits the carbon intensity of wherever they are drawn: an $18.8\times$ range between Algeria’s grid and Norway’s for identical computation and an identical guarantee. The premium came out the same on both datasets despite a fivefold difference in energy per round, so the ratio belongs to the privacy mechanism rather than the workload. The point is sharper for federated learning than for centralised training, because the computation sits on whatever grids the participating devices do and no operator chooses them.

E. Two Ways to Measure a Privacy Mechanism Wrongly

Instrumentation attached to a privacy mechanism tends to scale with the privacy parameter, so its cost mimics the effect under study: the accountant’s calibration search, run inside the federated loop, reproduced our entire observed ϵ trend while step counts stayed flat, and nothing in the resulting numbers announces it. The second hazard is an accounting error rather than a measurement one, and it survived into our own first draft. Both are generic to client-level DP in simulation, so we recommend four controls: derive noise multipliers before measurement begins, verify per round that executed and calibrated step counts agree, randomise run order, and recompute each client’s spent budget from logged participation rather than reporting the target.

F. Threats to Validity

Clients are virtual and share one GPU, so the ratios transfer to real devices but the absolute joules do not; confirming them on physical hardware is the immediate next step. Measurement is GPU-only and communication energy is modelled — for a 0.15–0.24 MB model over tens of rounds we expect it small, but have not shown it. One CIFAR-10 run is excluded for being measured in a different session, leaving that condition on two seeds. And our two datasets bound the effects rather than establishing them: the stopping round appears on one and not the other, and the tuning result inverts between them, so where the boundary falls is a question this work poses rather than settles.

VI. CONCLUSION

We measured what differential privacy costs in energy when added to federated learning, using hardware counters around each round on two datasets, with UCI-HAR partitioned so each client holds one study participant's data.

The per-round cost is power times time. The power factor is $1.19\times$ on both datasets to within half a per cent, despite different models and a fourfold difference in size — the arithmetic of per-sample gradients and clipping, apparently a hardware constant. The time factor is not transferable, and their product, $+51\%$ and $+34\%$, carries no systematic budget dependence. Everything that does acts through the round count, and it is the larger term: reaching 0.70 accuracy on HAR costs $5.7\times$ the non-private energy with only $\epsilon \geq 4$ arriving at all, while 0.85 is unreachable at any expenditure.

Three rules follow. Derive the round budget from the privacy budget rather than the task, since rounds consume budget and training to convergence is self-defeating. Choose the privacy budget and the accuracy requirement together, because a budget implies a ceiling no computation reaches. And check whether the private model is noise-limited before tuning it: the same two knobs that move nothing on HAR move accuracy by a factor of two on CIFAR-10. Paid in joules, the cost also inherits the local carbon intensity — an $18.8\times$ range between Algeria's grid and Norway's for an identical guarantee, on both datasets — which a privacy-utility evaluation cannot represent.

We also report two hazards. Instrumentation attached to a privacy mechanism scales with the privacy parameter, so its cost mimics the effect under study. And calibrating the noise against expected client participation under-charges the clients sampled most: all 29 of our private runs spend more than their stated budget, by up to 22%. Deriving noise multipliers before measurement, verifying step counts per round, randomising run order and recomputing each client's spent budget from logged participation are cheap controls for any such study.

Clients are virtual and share one GPU, so the ratios transfer but the absolute joules do not, and our two datasets bound the effects rather than establishing them: a stopping round is resolvable on CIFAR-10 below $\epsilon = 2$ and on UCI-HAR nowhere. What sets that boundary — task difficulty, the magnitude of the noise, or their ratio — is the question this work leaves open.

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